

# Manankumar Thakkar

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## SUMMARY

Software Engineer with 4+ years of experience designing scalable, distributed systems in Java and Spring Boot and building generative AI agents and LLM tools. Skilled in design patterns, reliability and scaling, multi-threaded performance, Kubernetes, and the full development life cycle. Proven in production: cut API latency **60%** and prevented a **6x data collision**.

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL, TypeScript, JavaScript  
**Backend:** Spring Boot, Microservices, Distributed systems, REST APIs, Concurrency, Kafka, Redis, Flask, Node.js  
**Frontend:** React, Next.js, Tailwind CSS, Three.js  
**AI / LLM:** Generative AI agents, LLM evaluation, RAG, MCP / tool use, LangChain, Calibration, Prompt engineering  
**Cloud & Data:** Kubernetes, Docker, GCP (BigQuery, GCS), Azure SQL, AWS, Terraform, Liquibase  
**SDLC & Ops:** CI/CD, Git, Code reviews, JUnit, Grafana, Prometheus, Splunk, SonarQube, Snyk

## EXPERIENCE

### Walmart Global Tech | Software Engineer III 02/2025 – Present

- Expanded advertiser booking flexibility **40%** by architecting the Vision Center feature in **Java/Spring Boot**, letting the Inventory Reservation Service set loop policy per location, and leading the move to seconds-based loop metrics.
- Cut API latency **60%** and **10,000+ database calls/day** by fixing an N+1 bottleneck with concurrent caching in **microservices**.
- Prevented a **6x cardinality collision** in **multi-tenant international expansion** by catching a key-grain design flaw across 7+ partner orgs and driving a durable country-scoped key over a regional patch.
- Built a **generative AI operations assistant** (Python, Flask, LangChain) on Walmart's internal LLM Gateway, giving engineers chat access to operational context via a REST API with session memory and response caching.
- Reduced production vulnerabilities **35%** by fixing **Snyk** findings, removing hardcoded credentials, and automating secret rotation.

### Walmart Global Tech | Software Engineer 07/2024 – 01/2025

- Extended operating-hours coverage to Deli, Bakery, and Service Deli departments across **4,000+ stores** by building **Java/Spring Boot** services and extending the store data model for Sam's Club Plus and Store Department Hours.
- Enabled targeted in-store ad placement for new retail formats (Grab-N-Go, Sam's Club Cafe) by building location-to-screen-group mapping and filtering across the stores, inventories, and device-commands **REST APIs**.

### George Mason University | AI Research Assistant 07/2023 – 07/2024

- Developed research tooling powered by **LLMs** (ChatGPT) that automated and streamlined manual research workflows.
- Provisioned and managed research infrastructure as code with **Terraform** for reproducible research environments.

### Nokia | Test Automation Engineer (Co-op) 09/2022 – 06/2023

- Reduced manual testing effort **50%** by building **Python test automation** suites for high-throughput systems.
- Cut defect density **20%** and improved code quality metrics **35%** by integrating automated tests into **Jenkins CI/CD** pipelines.

### Let It Wag | Full Stack Developer 06/2019 – 05/2020

- Improved backend efficiency **20%** while building **Node.js** services and REST APIs that handled real-time data for **13K+ users**.
- Delivered **90%** of projects on time while building a responsive **React** UI supporting 100+ user journeys.

## AI PROJECTS

### MCP Security Observatory | *Python, tree-sitter, GitHub Actions* | [GitHub](#)

- Engineered a static scanner that audits **MCP servers** without executing them, covering **1,600+ servers** from a 21,000+ repository corpus with 5 tree-sitter taint rules (shell exec, path traversal, tool-description injection, hidden Unicode, scope).
- Enforced responsible disclosure in code, withholding **800+ high/critical findings** until maintainers are notified; shipped a nightly pipeline, public dashboard, SARIF output, and **782 tests** with ruff, mypy, and CodeQL in CI.

### Escalation Gate | *Python, LLM evaluation, RAG* | [GitHub](#)

- Designed an evaluation harness for a **RAG answerability gate** on 600 SQuAD 2.0 items with a contamination control, showing test-set construction alone swung accuracy **30.5 points** (100% vs 69.5%).
- Measured calibration error **0.047** against a 0.039 noise floor and a routing band that skips generation for **14.5%** of passages; manual review found 6 of the 10 most confident "failures" were benchmark label errors.

## EDUCATION & CERTIFICATIONS

### George Mason University, Fairfax, VA – *Master of Science in Computer Science* 08/2021 – 05/2023

**Coursework:** Data Structures and Algorithms, Software Modeling and Architectural Design, Data Mining, Advanced NLP

- Walmart Applied Agentic AI: Foundation to Production (completed with capstone)